

Last Time Practice Series

Digital Electronics Part-1

By Sanjay Rath Sir



ABOUT ME

ESE

- ❖ Teaching Experience: 21+ Years
- ❖ Achievements:
 - AIR-16 (ESE 2005), AIR-21 (ESE 2008)
 - Mentored More Than 1,00,000+ Students
- ❖ Area of Expertise: Network Theory, Digital Circuits, Control Systems, EDC, Analog Circuits, Advanced Electronics

UPSC



SANJAY RATHI

8:00pm

MASTER MSQ

✓

SERIES

Expected MSQ Questions & Their Approach

EE/ECE/IN/CS

LIVE

from  **17th January**

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How to become an **IES Officer** in First Attempt?

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17th Jan



7:30 PM

Class-1



Ashutosh Saxena

M.Tech, IIT BHU
13+ Years of Experience



MAHA MARATHON

8 HOURS

Engineering Mathematics

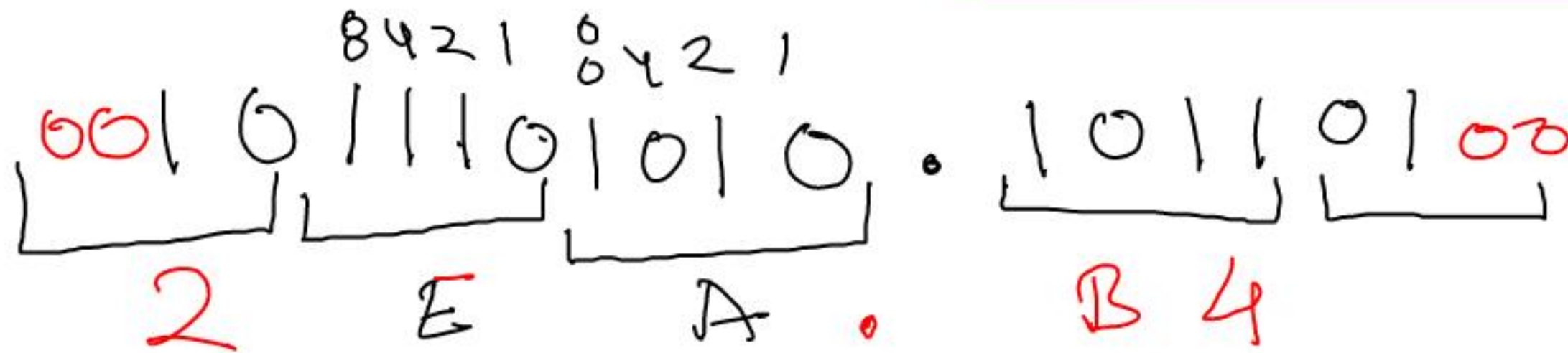
📅 21 Jan & 22 Jan | ⌚ Time: 10 AM

Session on App



- ① No. system
- ② Boolean Algebra
- ③ GATES
- ④ Adder
- ⑤ Decoder
- ⑥ multiplexers

Q. $(1011101010.101101)_2 = (X)_H$, then value of X is _____.



2EA.B4 Ans

- A — 16
- B — 11
- C — 12
- D — 13
- E — 14
- F — 15

mcq.

Q. Given that $(131)_x + (153)_x = (356)_x$. Then the value of base 'x' is _____.

A. 6

B. 2

C. 1

☒ D. None

$x > 6$

$$(1x^2 + 3x^1 + 1x^0) + (x^2 + 5x^1 + 3x^0) = 3x^2 + 5x + 6x^0$$

$$2x^2 + 8x + 4 = 3x^2 + 5x + 6$$

$x < 6$
 $x > 6$

$$1 \times 6^2 + 3 \times 6^1 + 5 \times 6^0$$

$$= 36 + 18 + 5 = \underline{\underline{59}}_{10}$$

$$(x7)_y = xy^1 + 7y^0$$

$$= (xy + 7)_{10} = 59$$

$$xy = 59 - 7 = 52$$

Q. Consider the equation $(135)_6 = (\underline{x7})_y$. The number of possible solutions is _____.

- A. 0
- ~~B. 3~~
- C. 2
- D. Infinite

$$y > 7$$

$$xy = 52$$

$$① \quad x = 1 \quad y = 52 > 7 \quad \checkmark$$

$$② \quad x = 2 \quad y = 26 > 7 \quad \checkmark$$

$$③ \quad x = 3 \quad 52/3 = \bar{y} \quad \times$$

$$④ \quad x = 4 \quad 52/4 = 13 = \bar{y}$$

$$⑤ \quad x = 5 \quad 52/5 = \bar{y} \quad \times$$

$$⑥ \quad x = 6 \quad y = 52/6 = 8.66 \quad \times$$

$$⑦ \quad x = 7 \quad y = 52/7 = 7.42 < \times$$

$$⑧ \quad x = 8$$

$$y = 52/8 = 6.5 < 7 \quad \times$$

Q. Find the Minimum Decimal equivalent of (1C5)

$$[1C5] = [2]_{10}$$

Hexa

$$\begin{aligned} [1C5]_H &= 1 \times 16^2 + 12 \times 16^1 + 5 \times 16^0 \\ &= 256 + 192 + 5 \\ &= (453)_{10} \end{aligned}$$

A	→	10
B	→	11
C	→	12
D	→	13
E	→	14
F	→	15

(4) Right.

(5) Wrong.

if $r = 10$

0 1 2 3 4 5 6 7 8, 9

$r = 11$

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A

$r = 12$

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B

$r = 13$

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C

$r > 12$

$r \geq 13$

$(1C5)_{13}$

$$= 1 \times 13^2 + 12 \times 13 + 5 \times 13^0$$

$$= 169 + 156 + 5 = \underline{\underline{(330)_{10}}}$$

$$\begin{array}{rcl}
 \underline{\underline{11001}} & - & 00111 \\
 & = & -7 \\
 \underline{\underline{1001}} & - & 0111 \\
 \\
 \underline{\underline{111001}} & = & - \underline{\underline{00111}} \\
 & & -7
 \end{array}$$

Q. 11001, 1001 and 111001 correspond to the 2's complement representation of which one of the following sets of numbers given below ?

A. 25, 9, 57

B. -6, -6, -6

✓ ~~C. -7, -7, -7~~

D. -25, -9, -57

$$\underline{\underline{11111001}} = -7$$

32 | 16 | 8 | 4 | 2 | 1

0 | 0 | 0 | 0 | 0 | 1

+17

→ 1 0 1 1 1 1 = -17

Q. 2's complement representation of -17 in 6-bit machine is _____.

A. 101110

B. 111110

☒ C. 101111

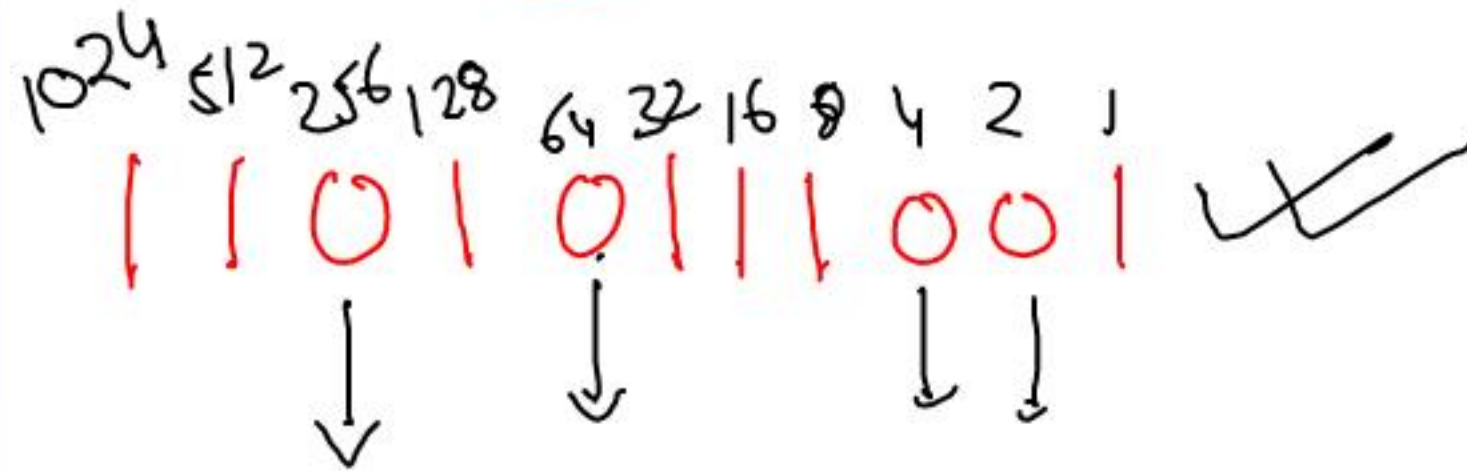
D. 110001

BCD (Binary Coded Decimal)



• $(\underline{1101011001})_{\text{BCD}} = (?)_{10}$

• $(\underline{11010111001})_2 = (?)_{10}$



$(\underline{1721})_{10}$

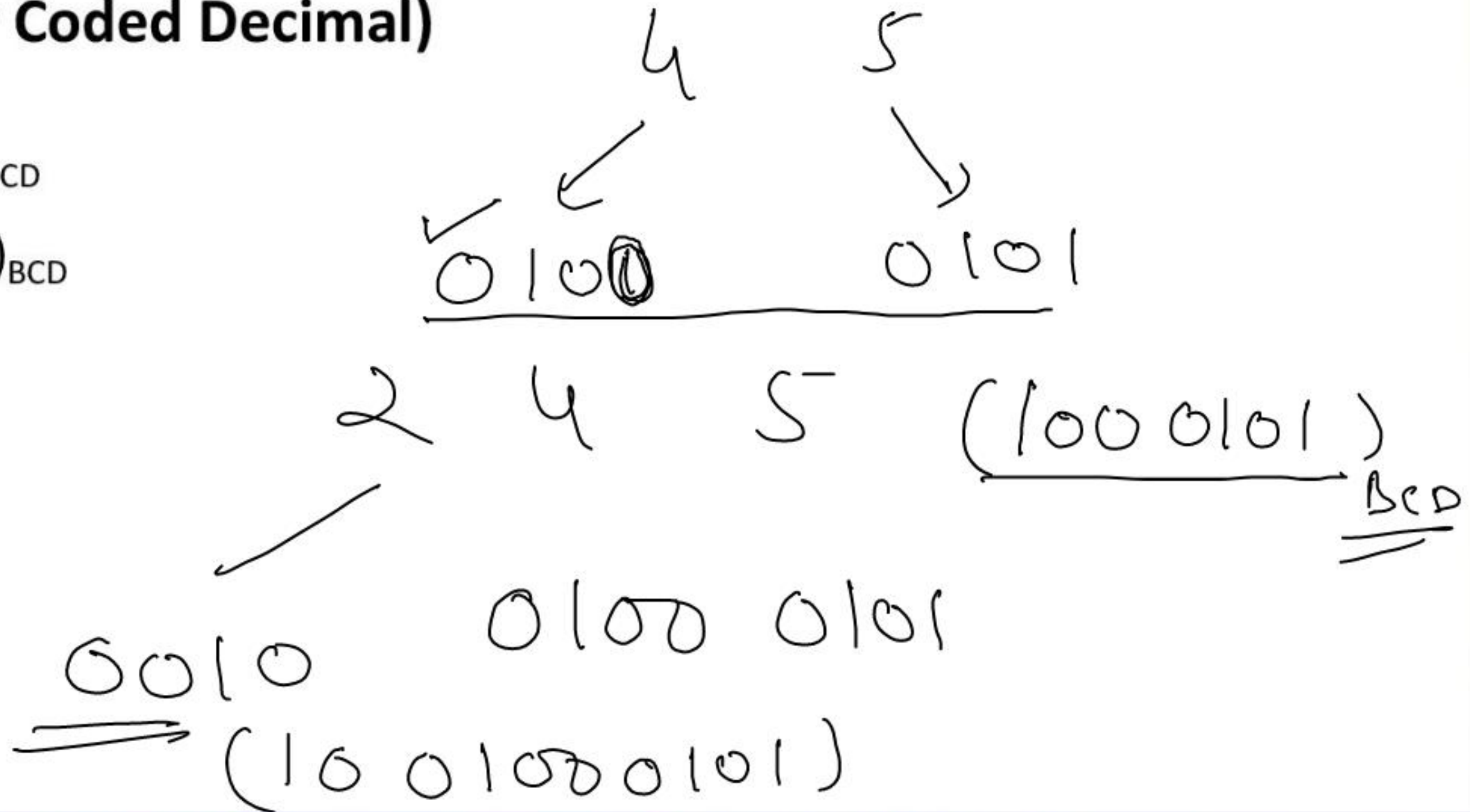
$| | | | | | | | | \rightarrow 2047$

$2^{11} = 2048$

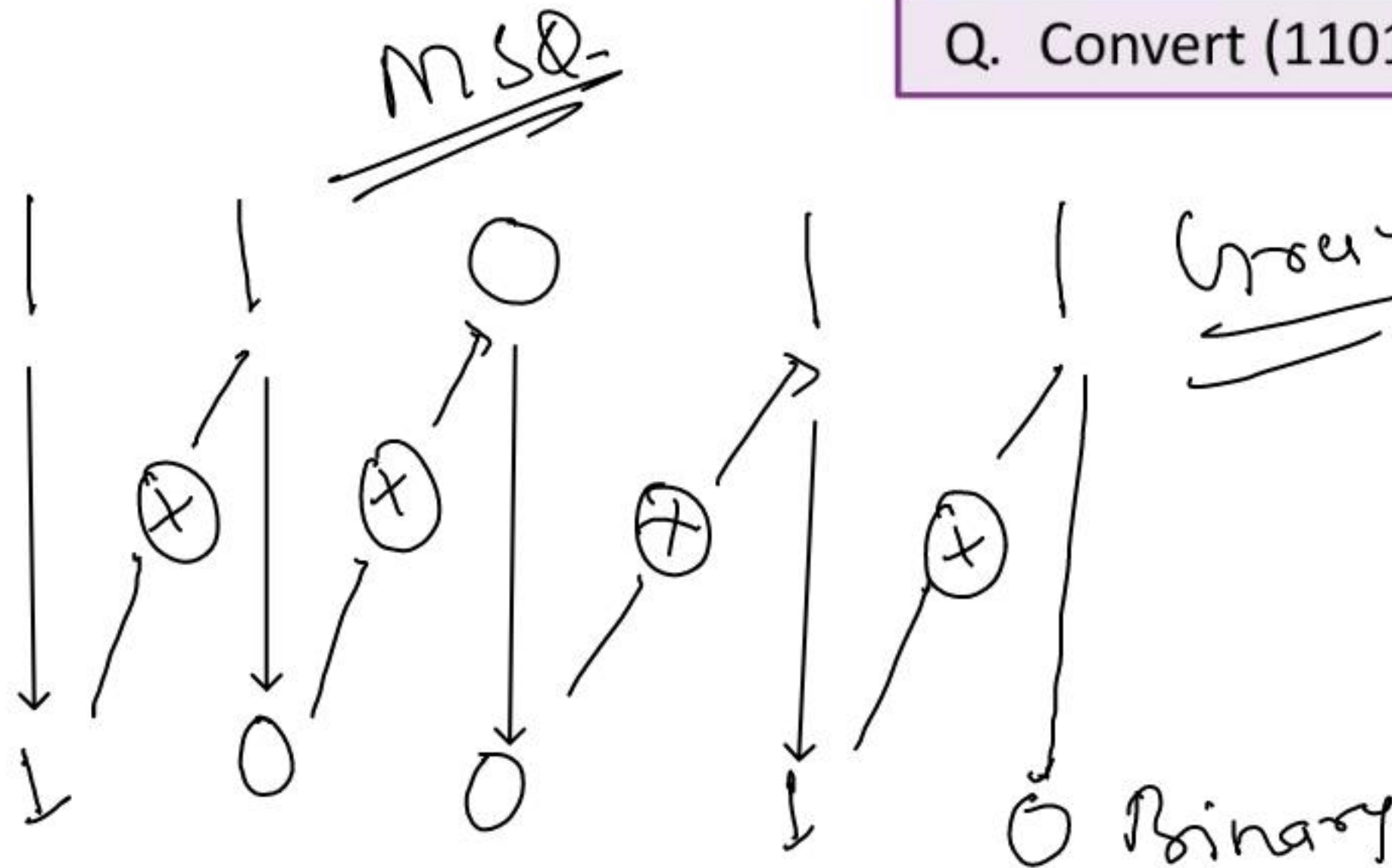
$0 \rightarrow 2047$

BCD (Binary Coded Decimal)

- $(45)_{10} = (?)_{BCD}$
- $(245)_{10} = (?)_{BCD}$



Q. Convert $(11011)_{\text{Gray}}$ to binary code?



Gray

8 PM
msb

$(10011)_{\text{Gray}}$

- (a) 10010 ✓
- (b) Reflecting ✓
- (c) Cyclic code ✓
- (d) Shift Encoding ✓

$$F(ABC) = \{ \}$$

Q. Determine the minterm of the function
 $F = A + B'C'$.

	A	B	C	F
0	0	0	0	1 ✓
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

$$F(ABC) = \{0, 4, 5, 6, 7\}$$

$$A(B+B') + (A+A')(B'C')$$

$$AB + AB' + AB'C' + A'B'C'$$

$$AB[C+C'] + AB'C(C+C') + AB'C'$$

$$ABC + ABC' + AB'C + AB'C' + A'B'C'$$

A	B	
0	0	0
0	1	1
1	0	1
1	1	0

$$X \oplus \bar{X} =$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1 = 1$$

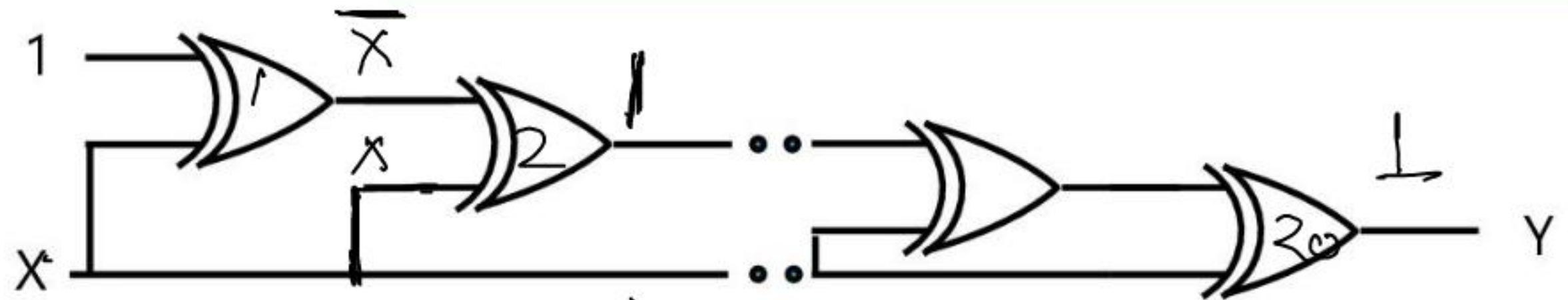
Q. If the input to the digital circuit consisting of a cascade of 20 X-OR gates is X, then the output Y is equal to

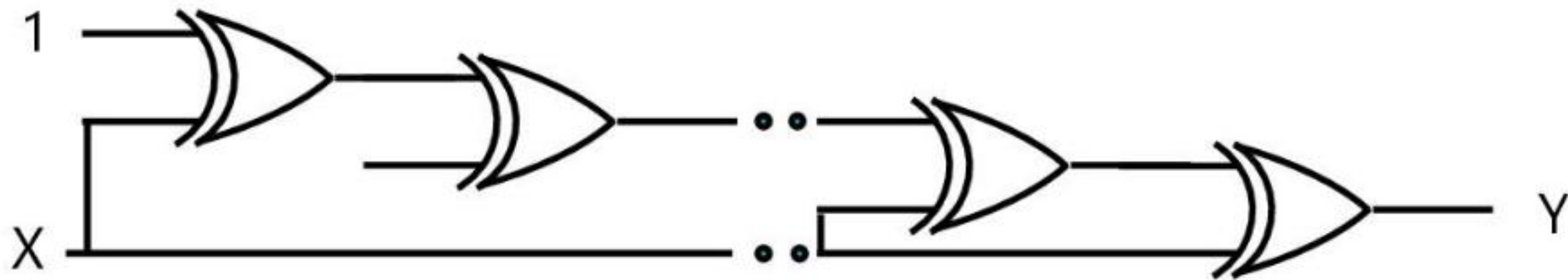
A. 0

B. X'

☒ C. 1

D. X





$$F = \overline{\overline{xy} \cdot \overline{yz}} \checkmark$$

$$= \overline{\overline{xy}} + \overline{\overline{yz}}$$

$$= xy + yz$$

$$y[x + z]$$

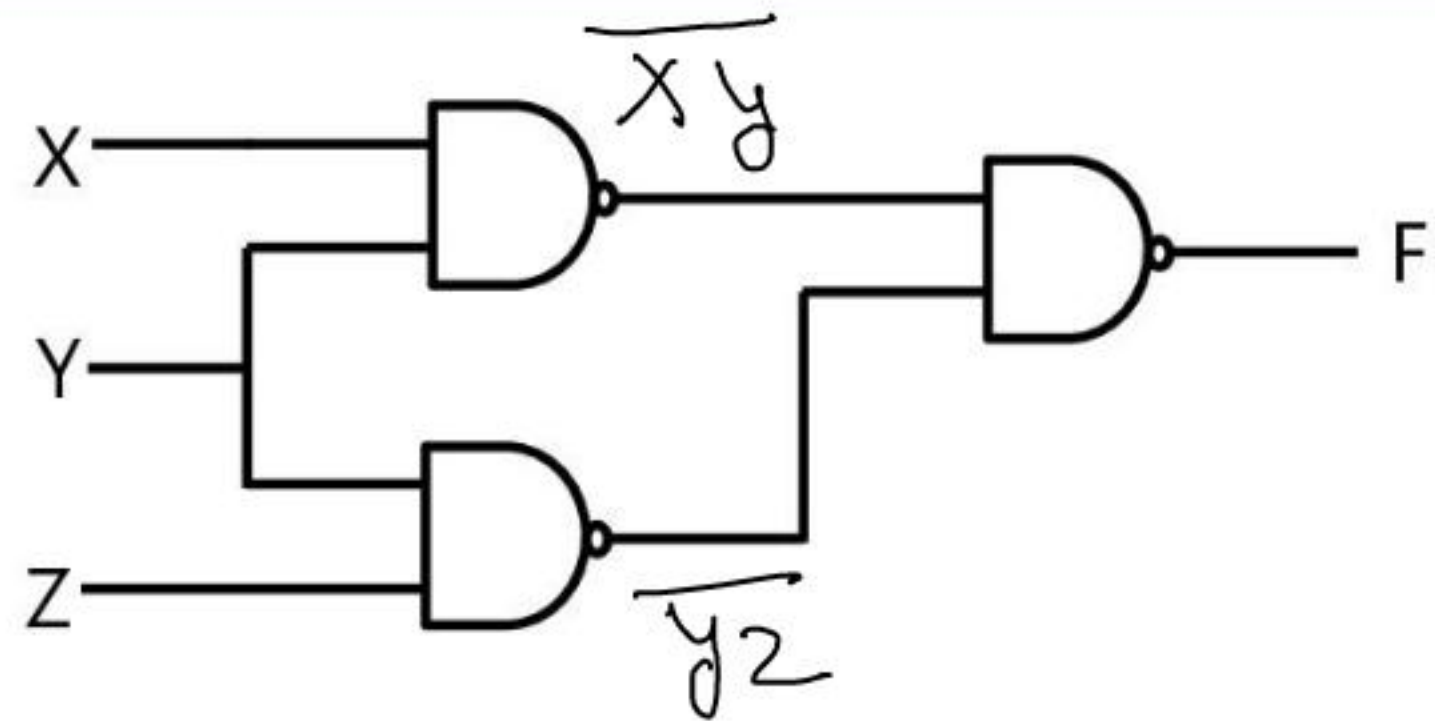
Q. Consider the following circuit.

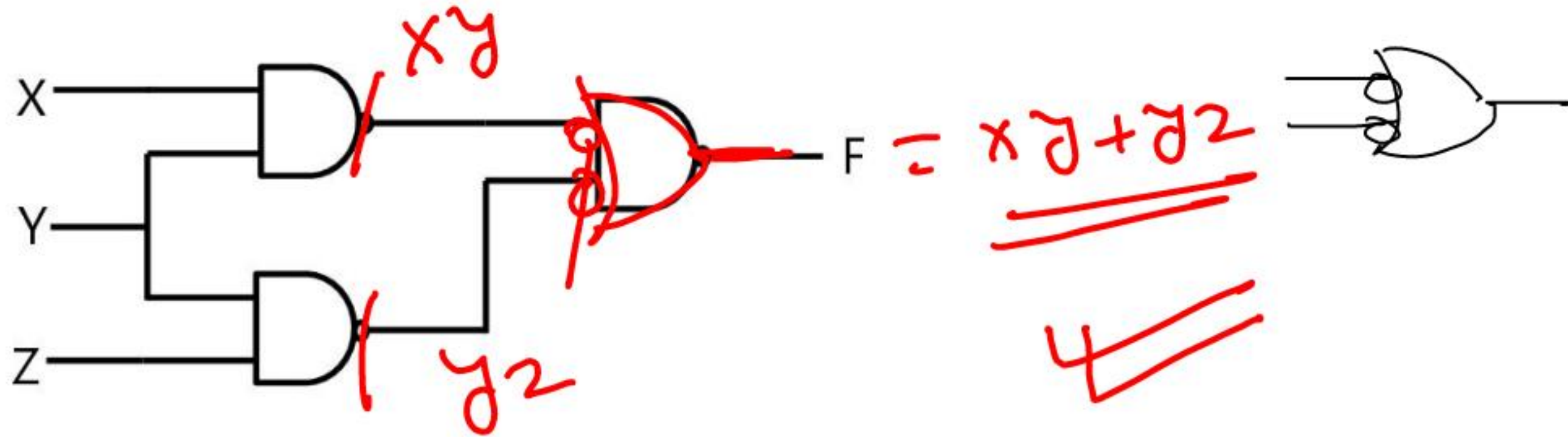
A. F is independent of X

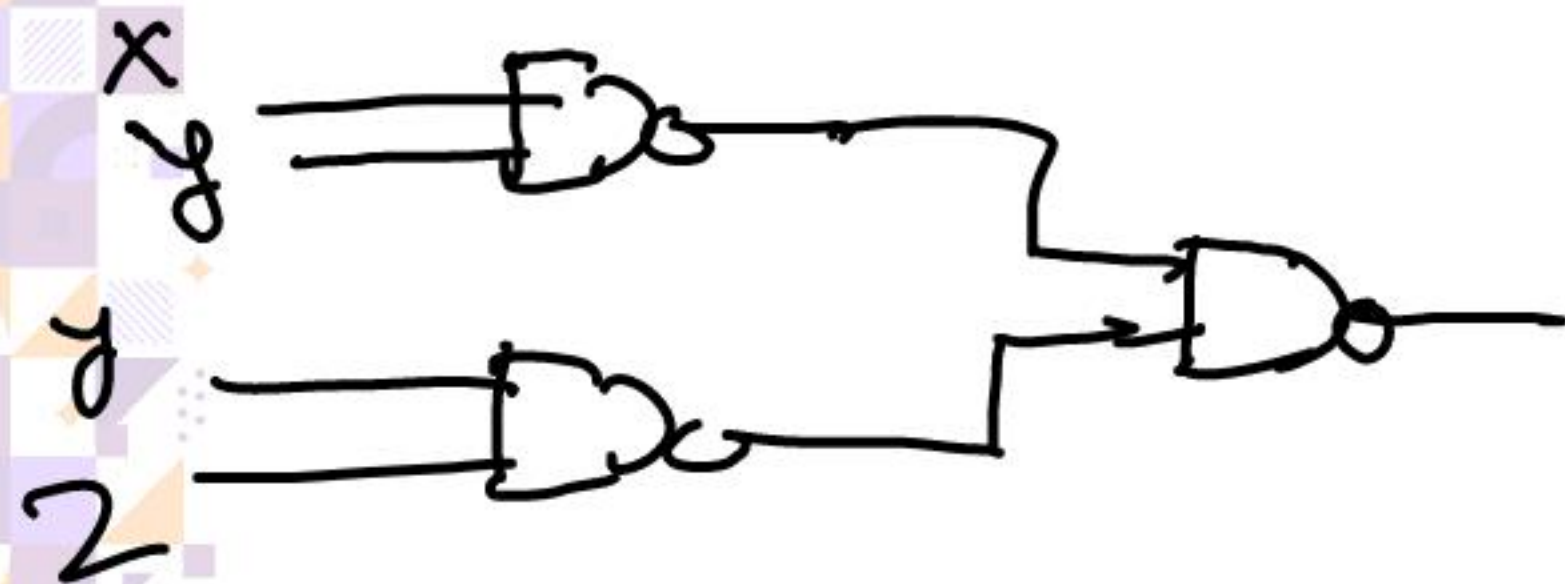
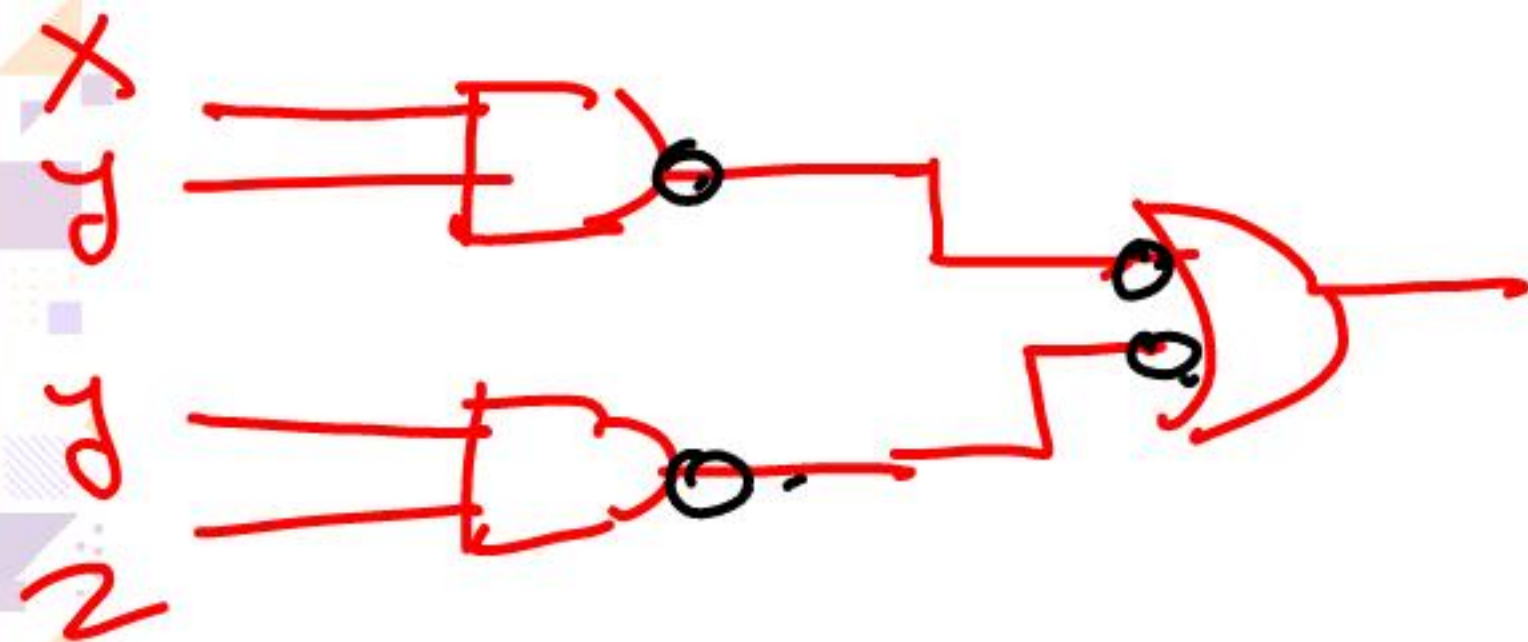
B. F is independent of Y

C. F is independent of Z

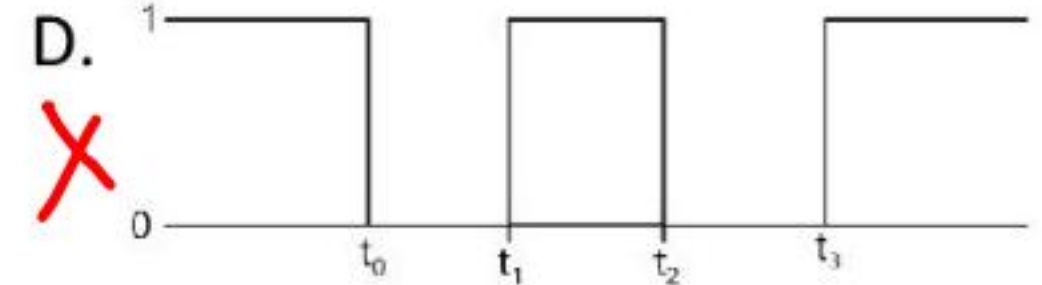
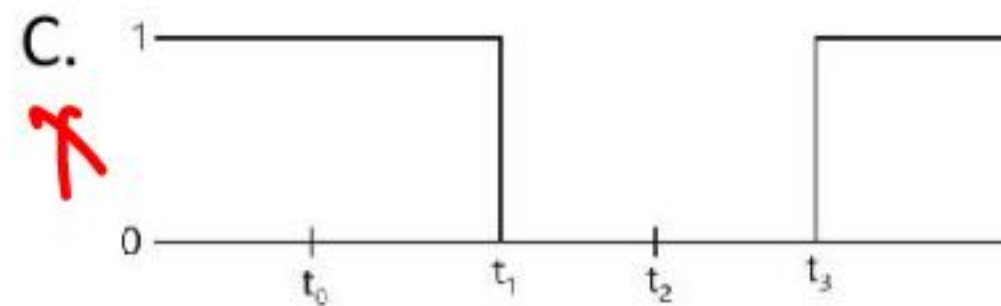
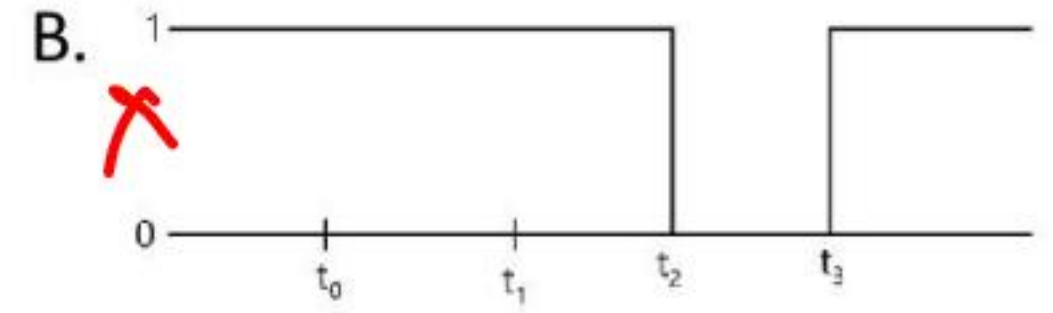
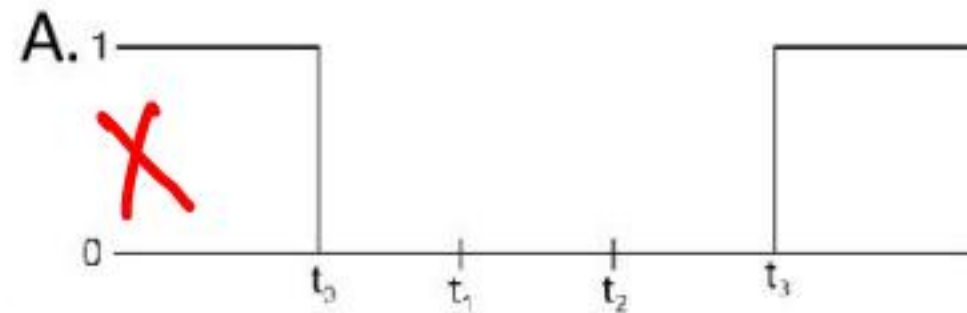
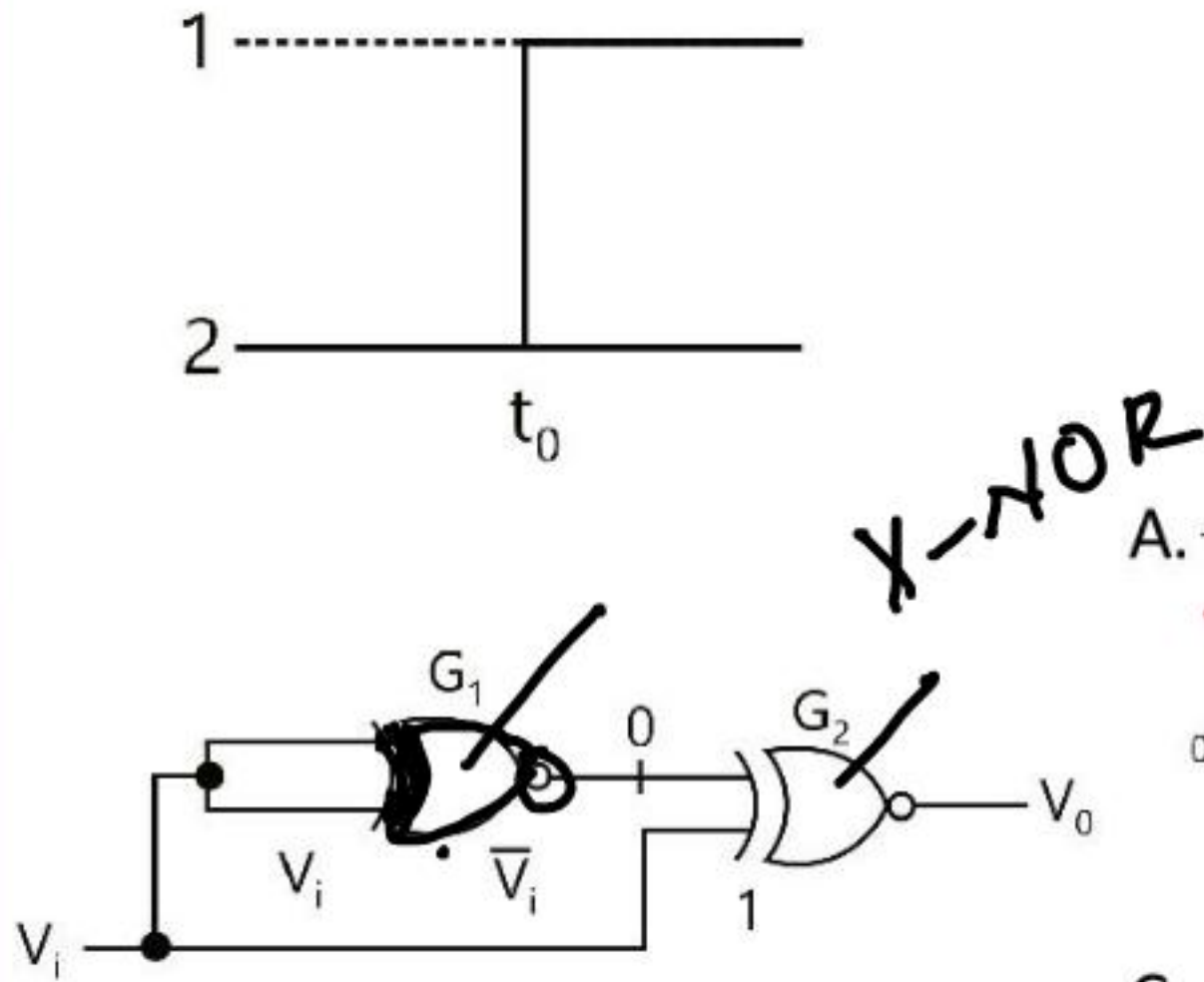
☒ D. None of the X, Y, Z is redundant

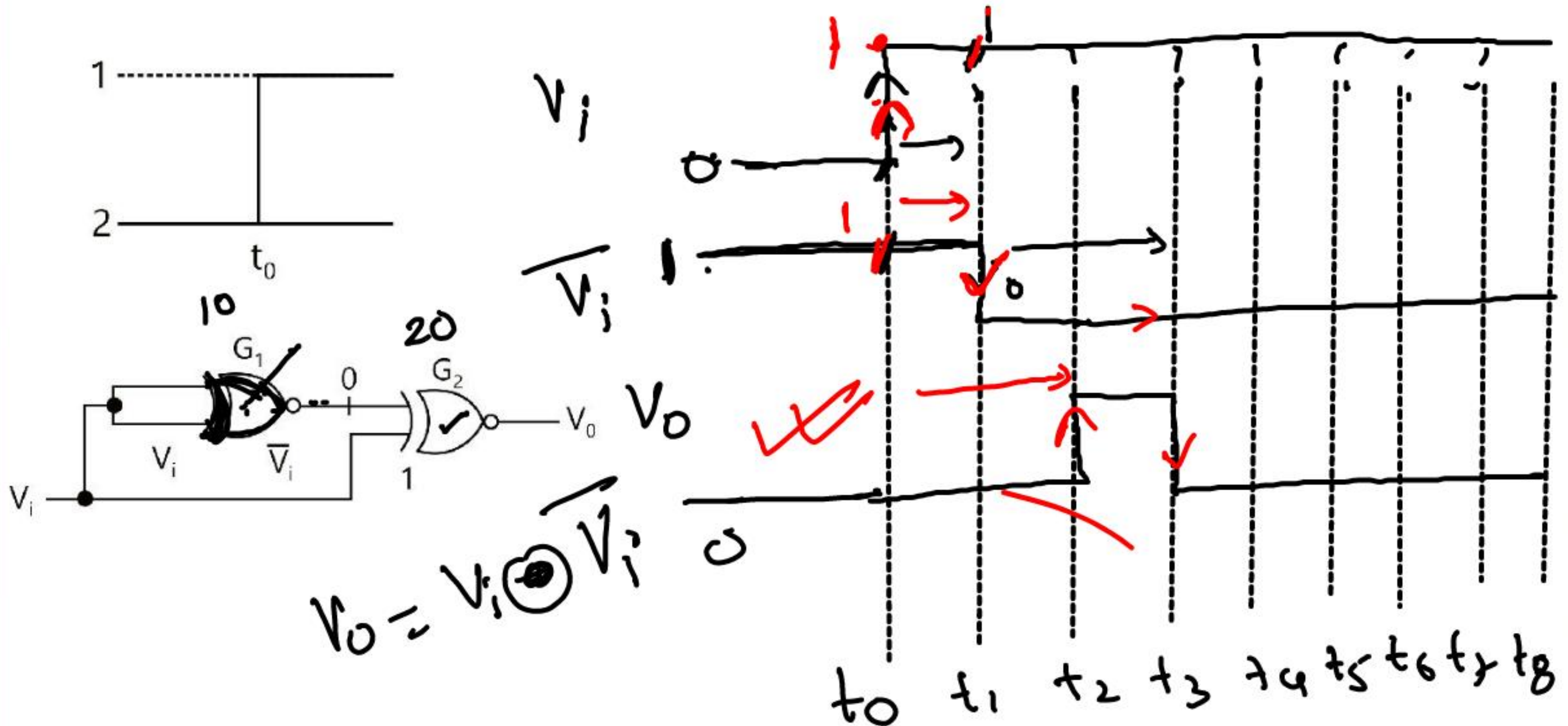






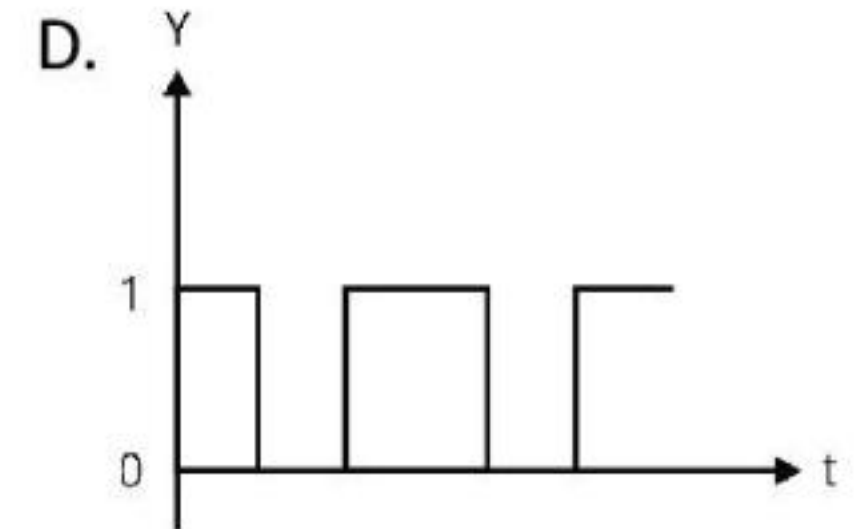
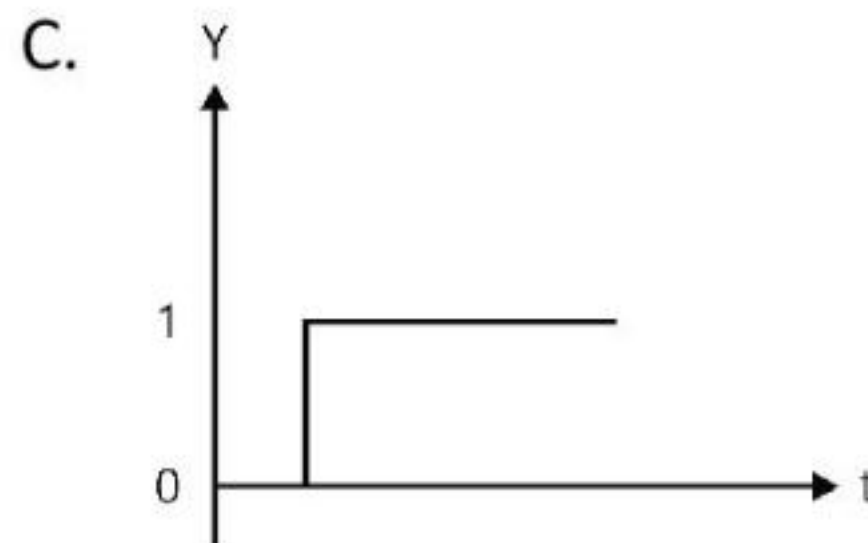
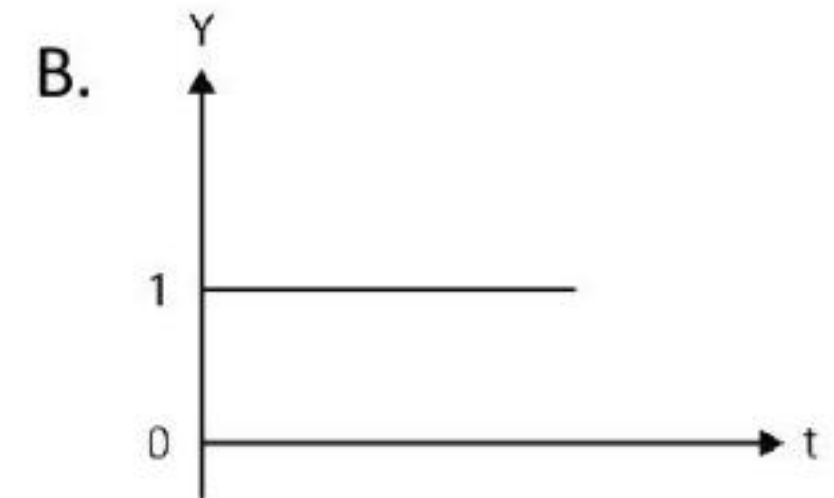
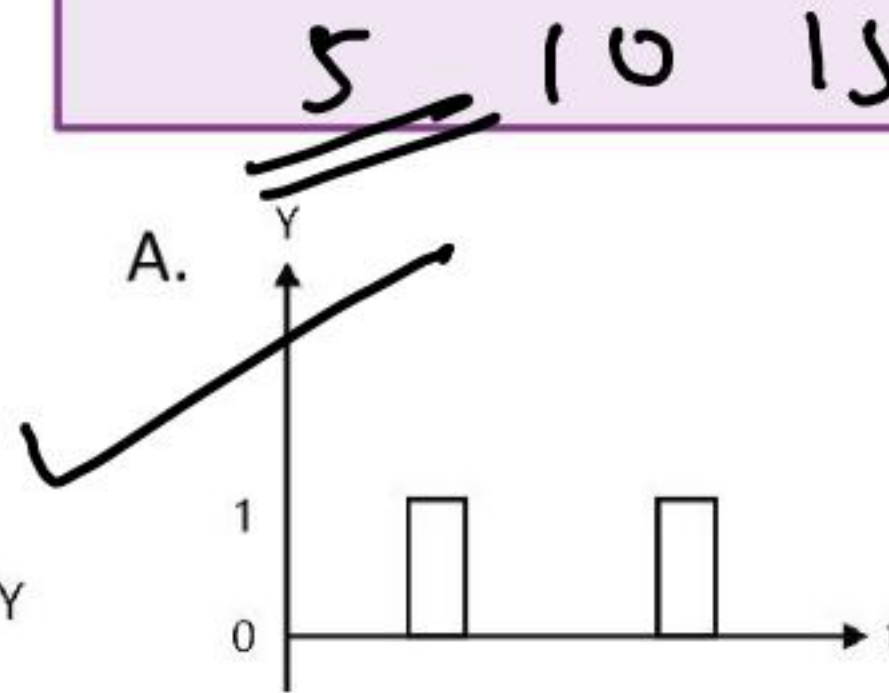
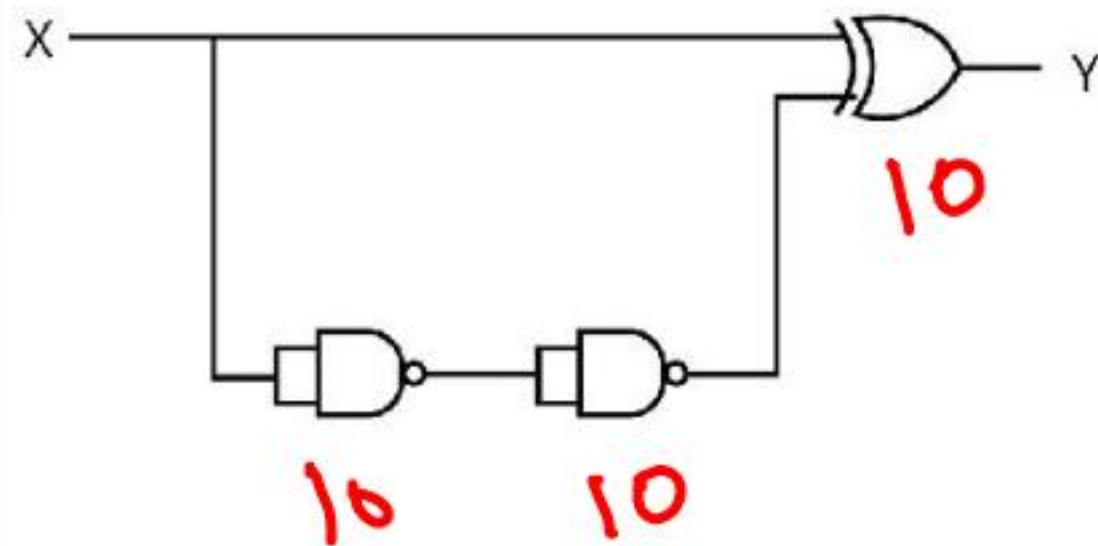
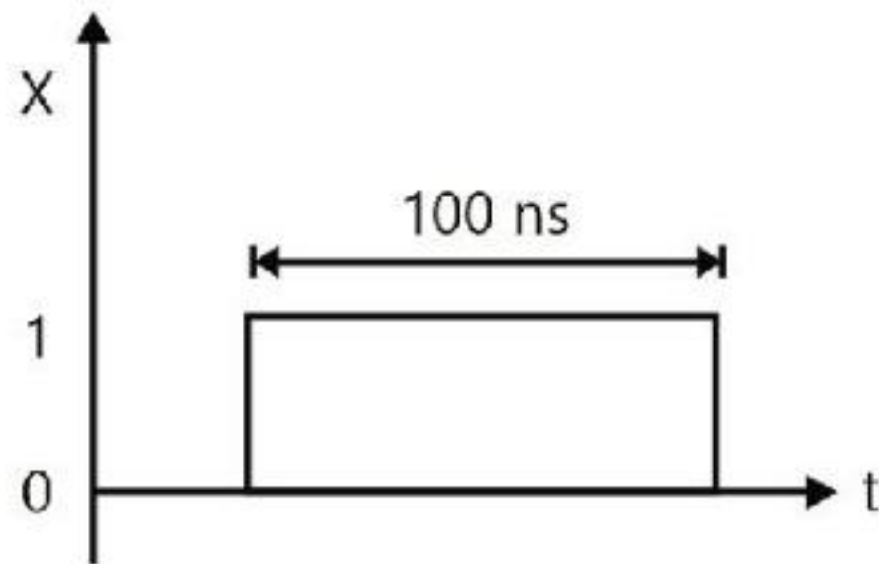
Q. The gates G_1 and G_2 in the figure have propagation delays of 10n sec and 20 n sec, respectively. If the input V_i makes an abrupt change from logic 0 to 1 at time $t = t_0$, then the output wave form V_o is

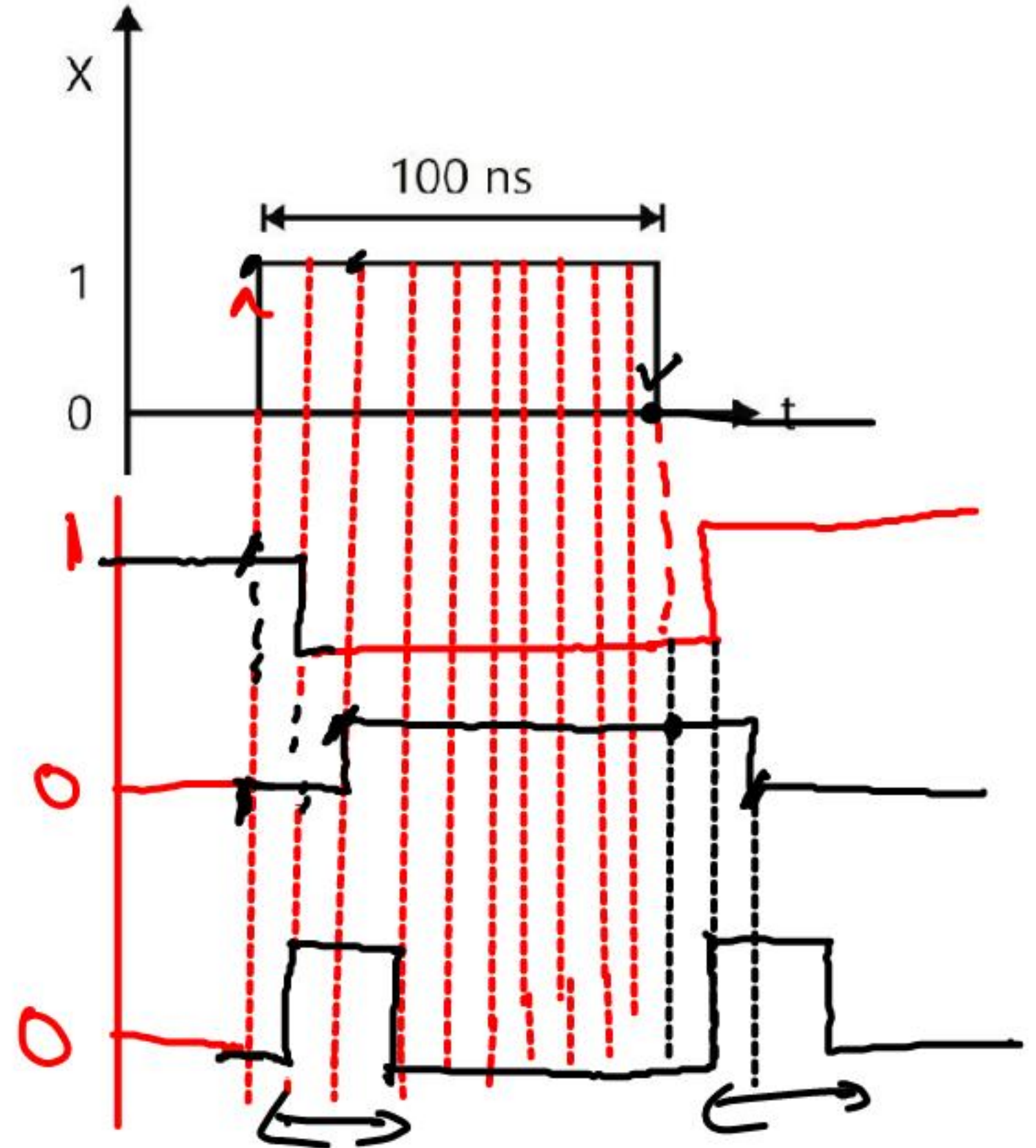
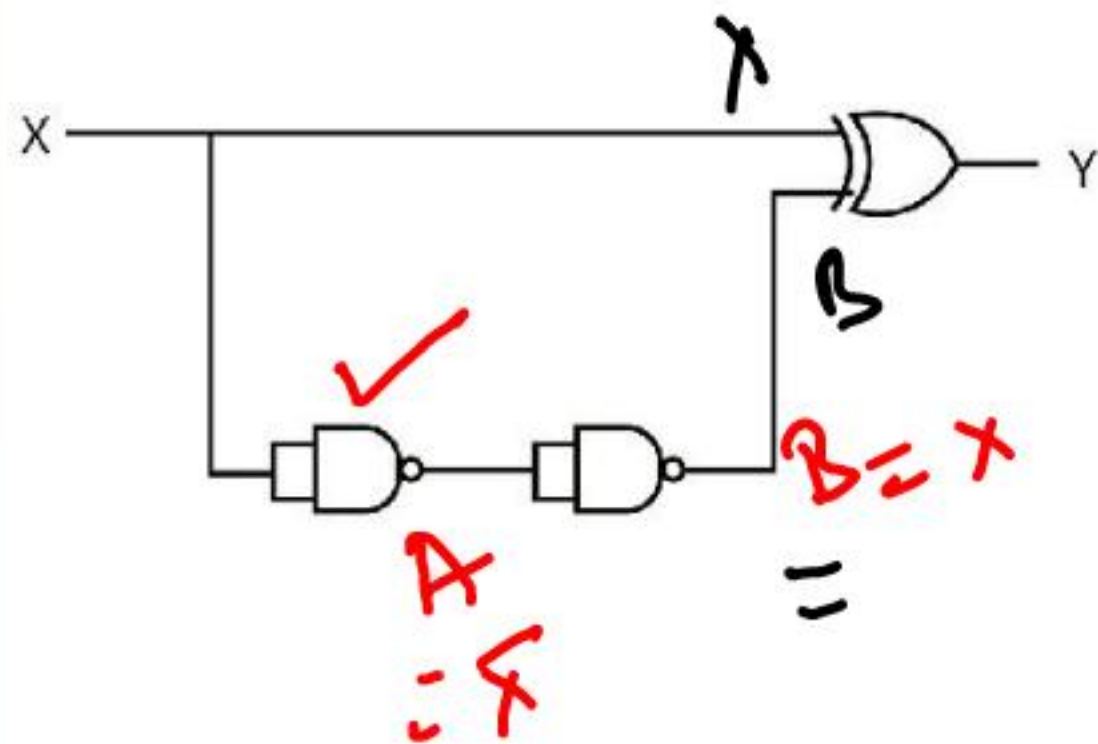




Q. The TTL circuit shown in the figure is fed with the waveform X (also shown). All gates have equal propagation delay of 10 ns. The output Y of the circuit is

[GATE 2010, 2 MARKS]





PQ \ RS	00	01	11	10
00	1			
01		1		1
11	1			
10		1		1

Q. Find the number of Prime implicants & Essential prime implicants in the given K

A. 4, 5

B. 0, 6

C. 6, 0

☒ D. 6, 6

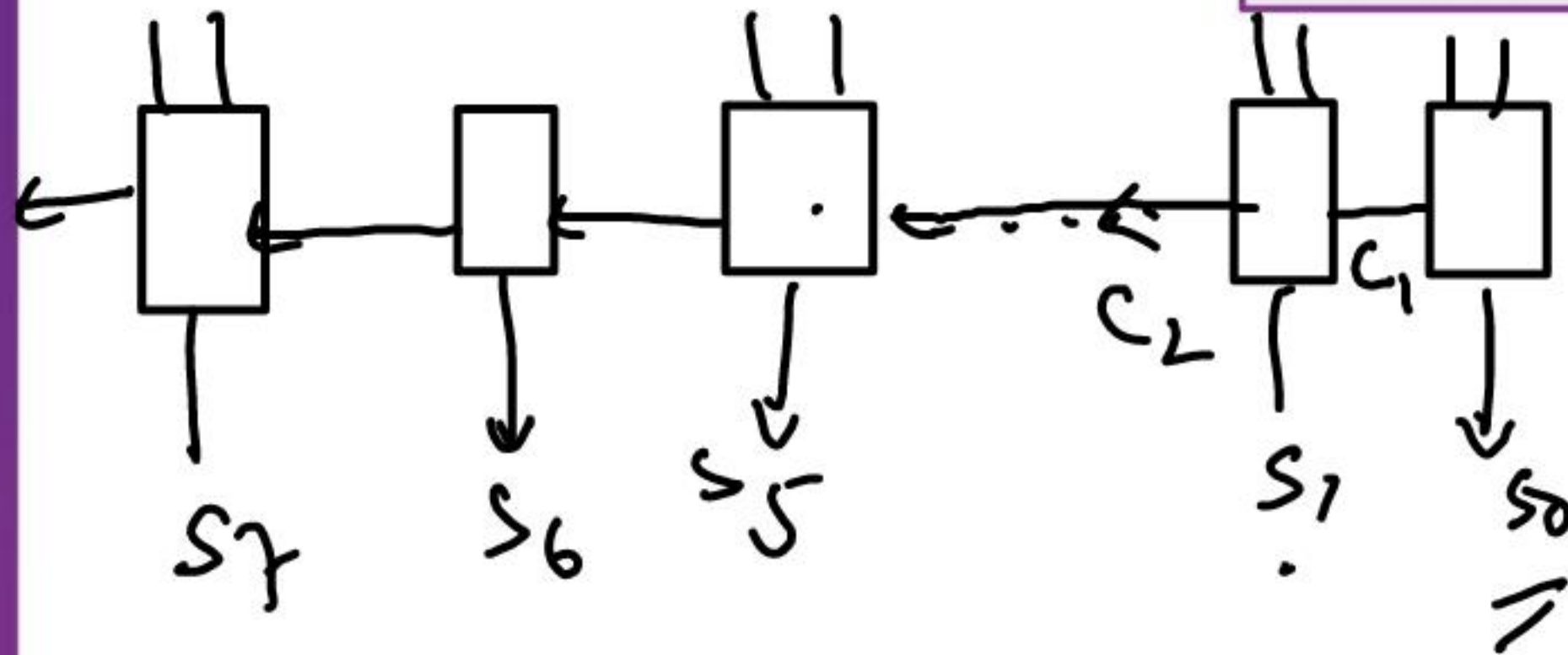
$$\bar{P}Q\bar{R}\bar{S} +$$

PI $\rightarrow$ 6
EPI $\rightarrow$ 6

$$\begin{array}{lcl} 2^0 & \rightarrow & 1 \\ 2^1 & \rightarrow & 2 \\ 2^2 & \rightarrow & 4 \\ 2^3 & \rightarrow & 8 \end{array}$$

$$\begin{array}{r}
 A_7 A_6 A_5 \quad A_0 \\
 B_7 B_6 B_5 \quad B_0 \\
 \hline
 C_8 S_7 \quad S_0
 \end{array}$$

Q. Consider an 8-bit ripple adder in which each full adder takes 15ns to generate the carry and 20ns to generate the Sum. What is the maximum rate of addition which can be obtained using this adder_____?



$\approx \frac{1}{\text{Delay}}$

8 M addition/sec

$$= (n-1) \tau_c + \max(\tau_s, \tau_c)$$

$$= (8-1) \times 15 + \max(20, 15)$$

$$= 7 \times 15 + 20$$

$$= \underline{\underline{125 \text{ nsec}}}$$

$$\text{No. of Additions/sec} = \frac{1}{\underline{\underline{125 \times 10^{-9}}}} = \frac{1000 \times 10^6}{125} = \underline{\underline{8 \times 10^6 / \text{sec}}}$$

$$\left. \begin{array}{l} \text{Combination} = 2^{2n} \\ (A=B) = 2^n \\ \left. \begin{array}{l} A > B \text{ or } \\ A < B \end{array} \right\} = \frac{2^{2n} - 2^n}{2} \end{array} \right\}$$

Q. For a 8-bit comparator the number of combinations for which binary number $A = B$ is _____.

A. 512

B. 64

✓ C. 256

D. 128

$$2^8 = 256$$

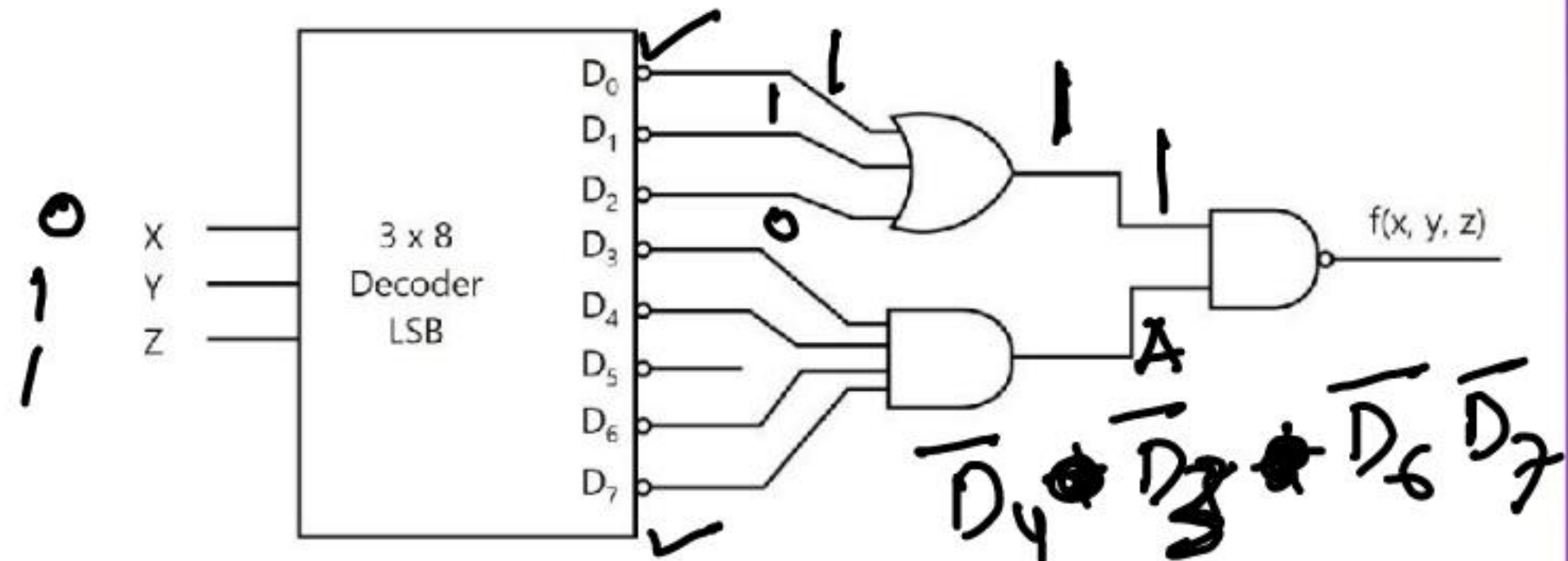
msd.

msd.

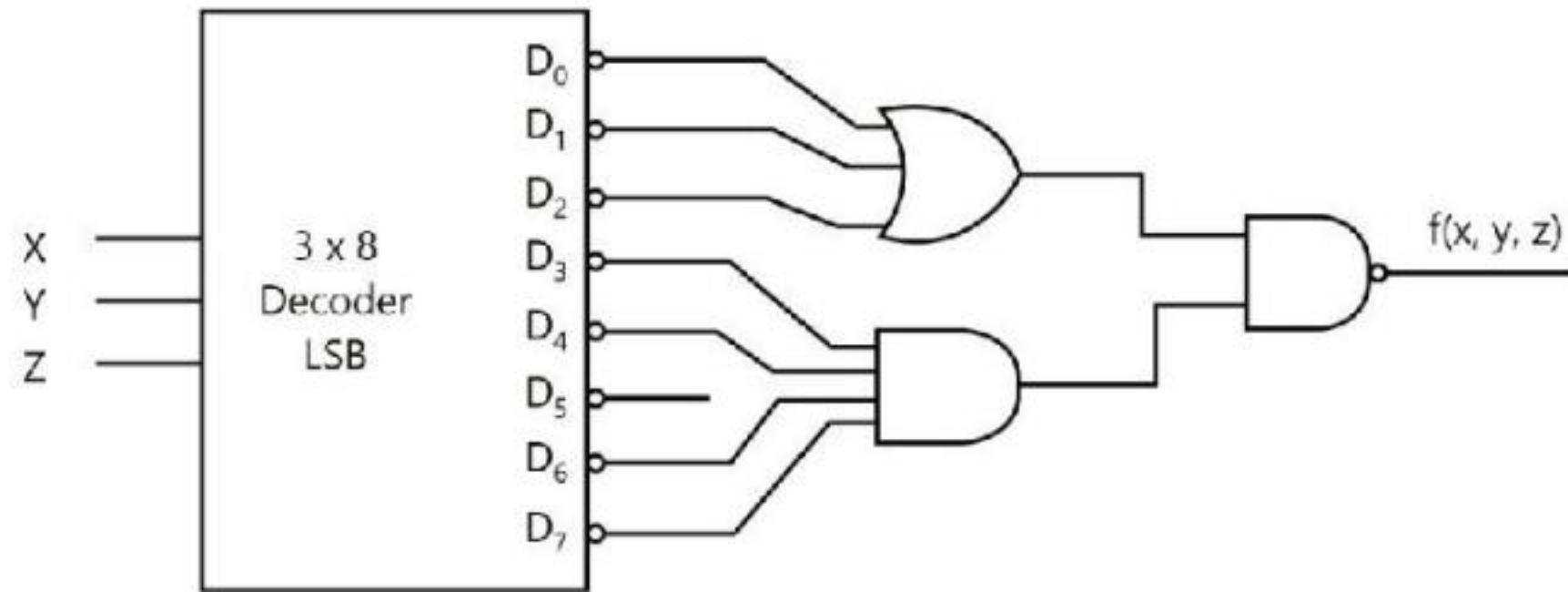
Inverted O/p

$D_7 D_6 D_5 D_4 D_3 D_2 D_1 D_0$
1 1 1 1 0 1 1 1

Q. Find the number of minterms in the function $f(x, y, z)$ for the decoder circuit shown below is _____.



$$f = \overline{D_4} \overline{D_3} \overline{D_6} \overline{D_3} = D_3 + D_4 + D_6 + D_7$$



How?

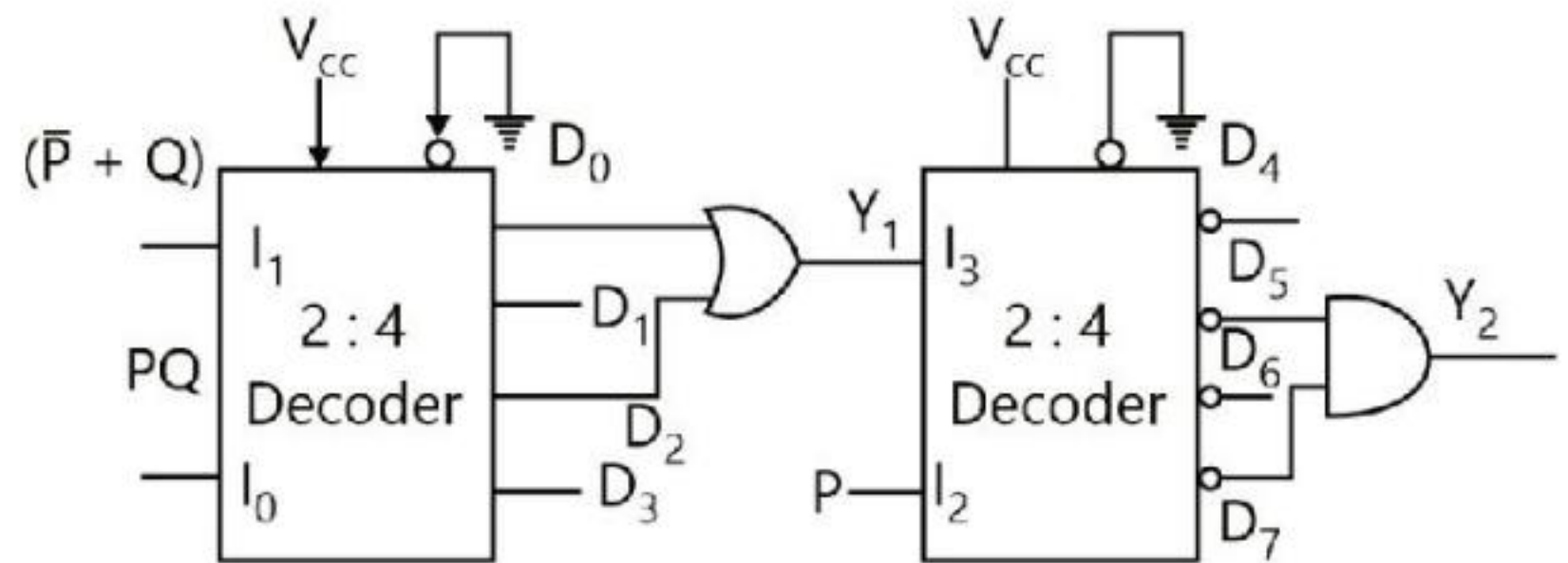
Q. From the given decoder circuit shown below, find output Y_1 and Y_2

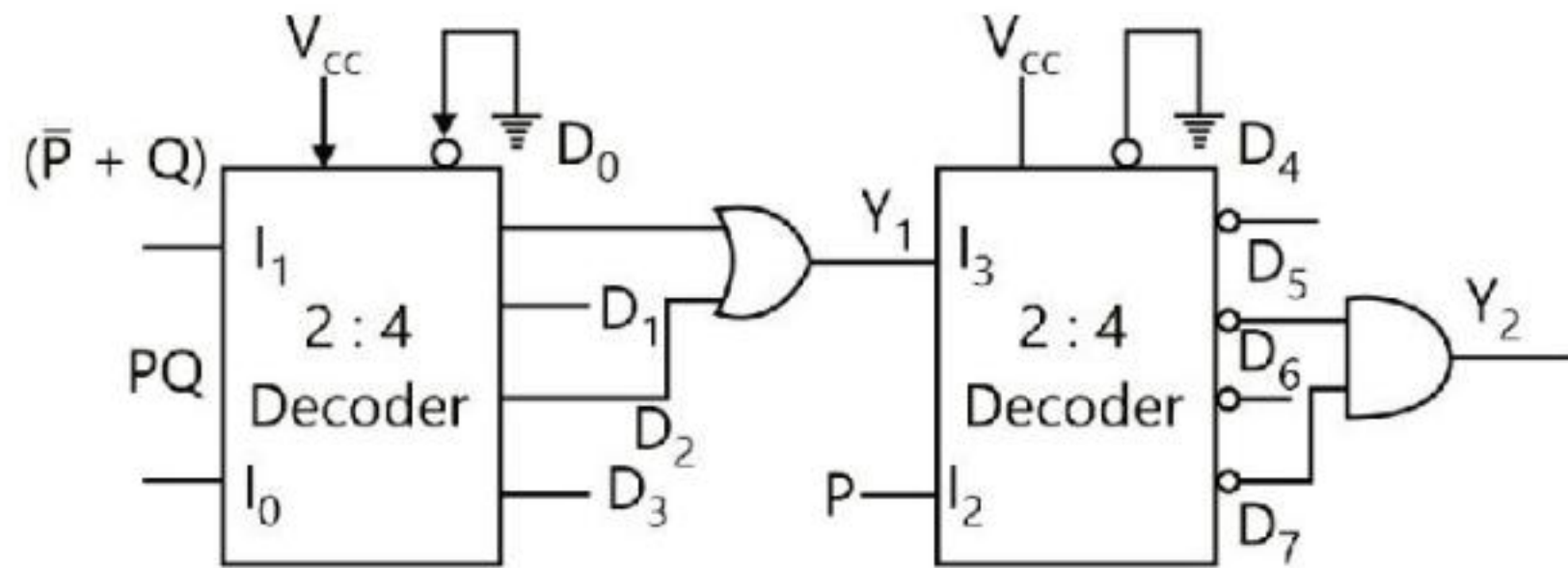
A. Y_1 is NAND of P , Q and $Y_2 = \bar{P}$

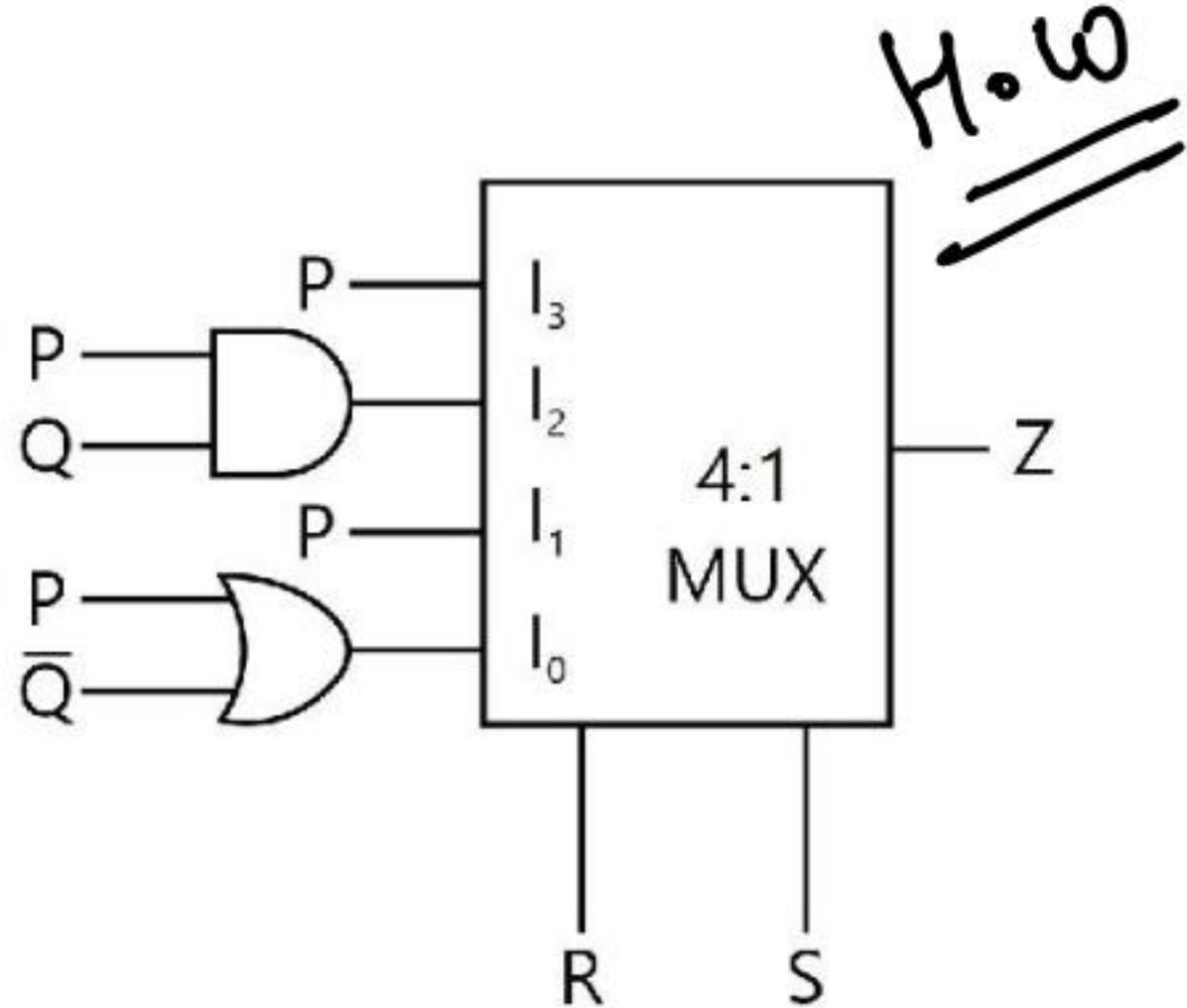
B. Y_1 is NOR of P , Q and $Y_2 = \bar{P}$

C. Y_1 is NAND of P , Q and $Y_2 = P$

D. Y_1 is NOR of P & Q and $Y_2 = P$





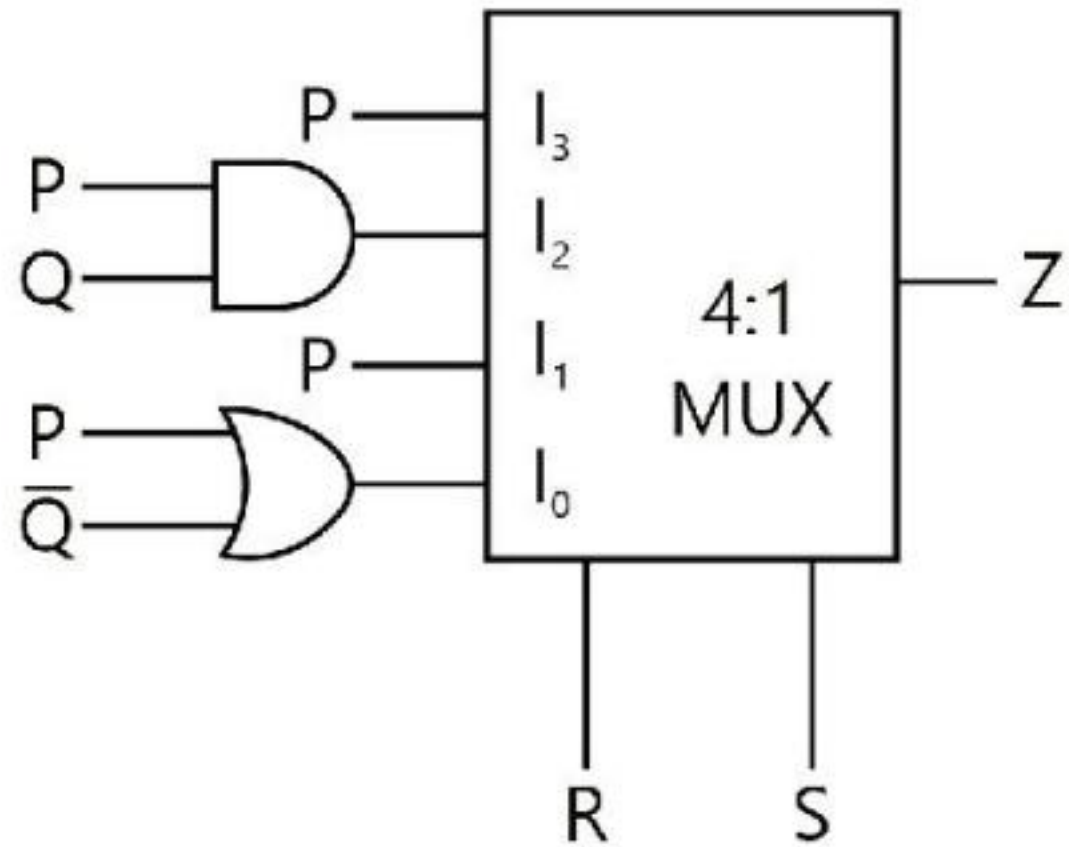


Q. For the circuit shown in the following figure, $I_0 - I_3$ are inputs to the 4:1 multiplexer. R (MSB) and S are control bits,

The output Z can be represented by

- A. $PQ + P\bar{Q}S + \bar{Q}\bar{R}\bar{S}$
- B. $P\bar{Q} + PQ\bar{R} + \bar{P}\bar{Q}\bar{S}$
- C. $P\bar{Q}\bar{R} + \bar{P}QR + PQRS + \bar{Q}\bar{R}\bar{S}$
- D. $PQ\bar{R} + PQR\bar{S} + P\bar{Q}\bar{R}S + \bar{Q}\bar{R}\bar{S}$

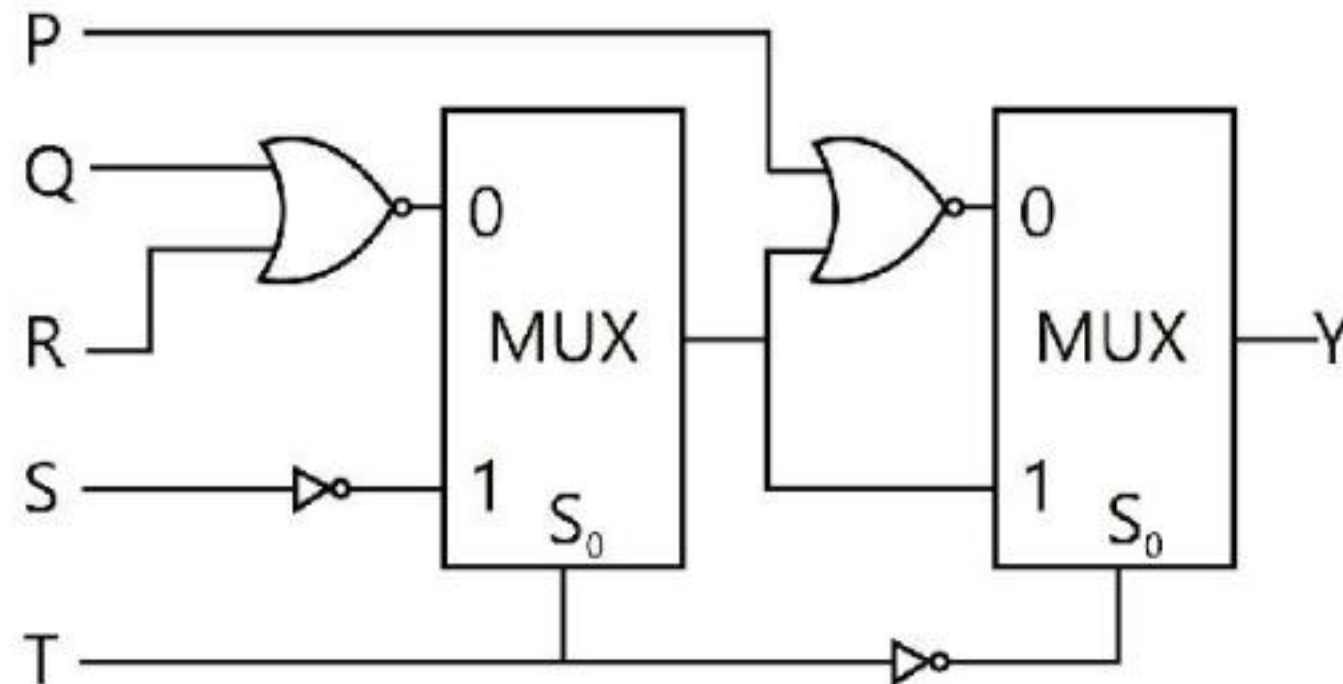
[GATE-2008, 2 MARKS]

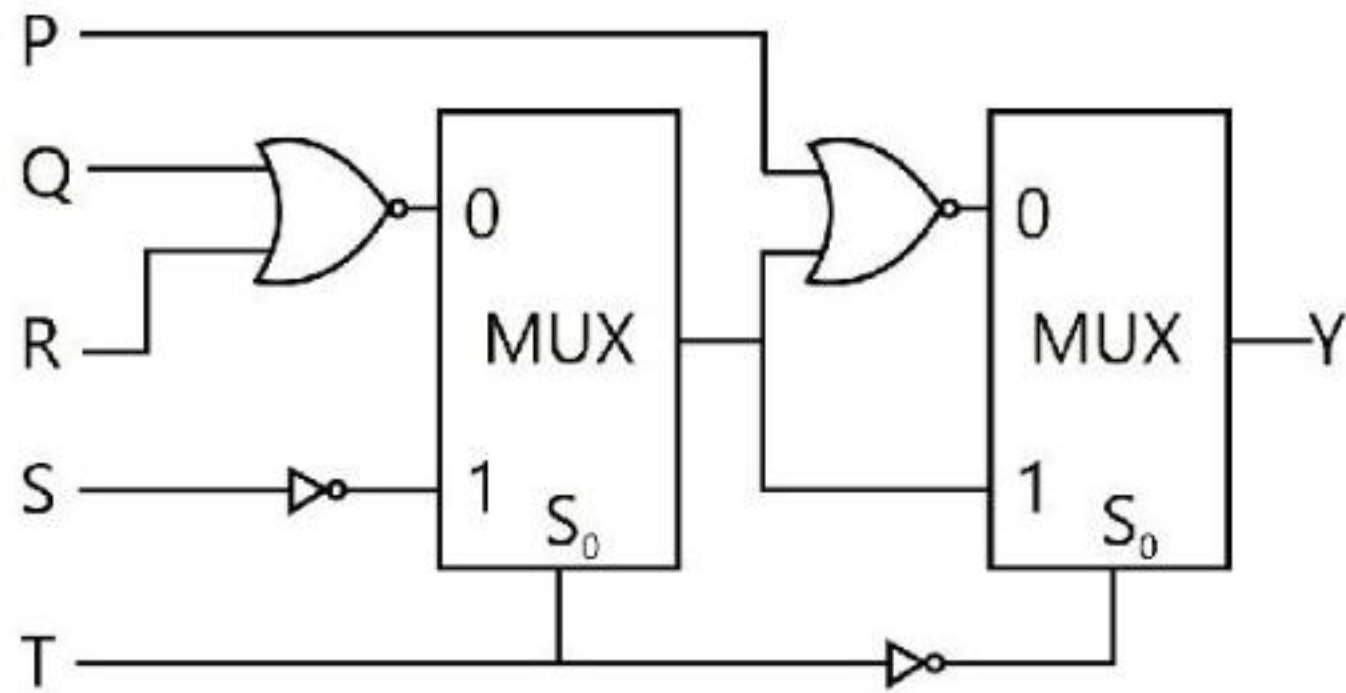


17.03

Q. For the circuit shown in the figure, the delays of NOR gates, multiplexers and inverters are 2 ns, 1.5 ns and 1 ns, respectively. If all the inputs P, Q, R, S and T are applied at the same time instant, the maximum propagation delay (in ns) of the circuit is _____.

[GATE-2016, 2 MARKS]





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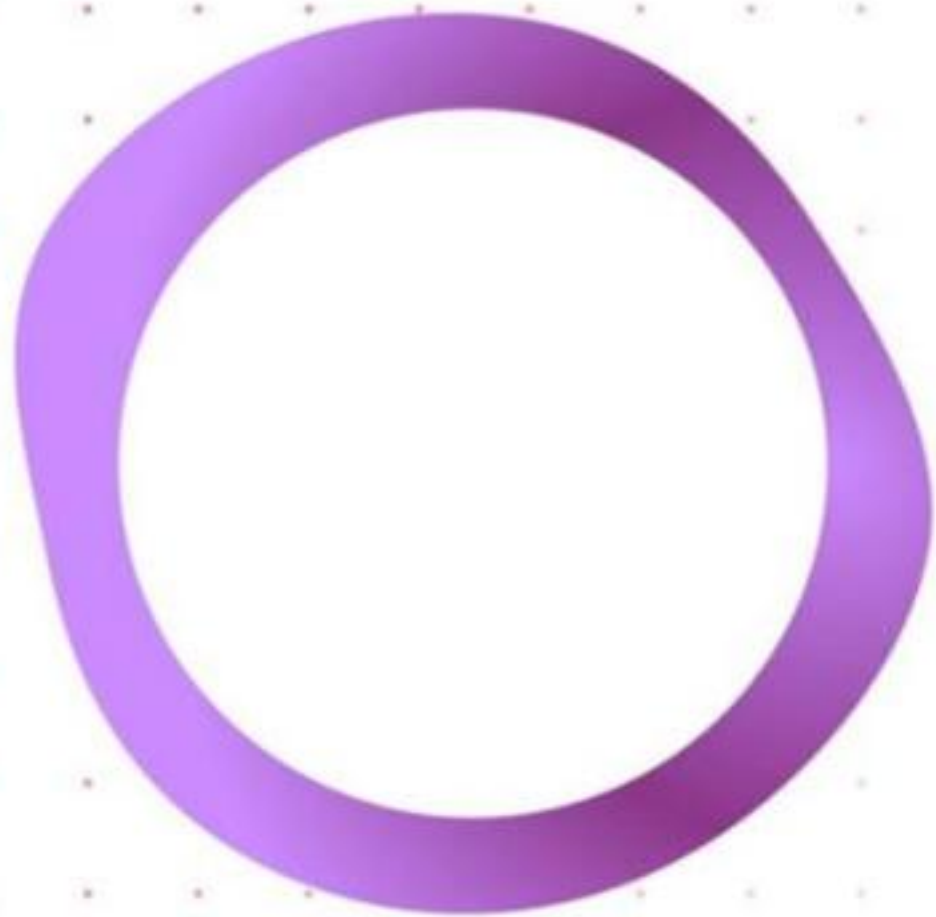
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